Chapter 1:
Overview of Databases

Concepts of a database system

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1. Introduce
2. File System
3. Define a database
4. Users of the database
5. Database management system
6. Levels of representation of a database

1. Introduction

- What is the definition of data?

Data are events, concepts, figures stored according to the purpose of use.

Data is described in many forms: characters, images, sounds...

2. File system

- Is a collection of individual files serving a purpose of the using unit.
- **Advantage** :
  - Rapid application deployment
  - Ability to respond quickly and promptly (because it only serves a limited purpose)
- **Disadvantages** :
  - Data duplication → waste, inconsistent data
  - High cost
  - Poor data sharing

3. Database (1)

- **Define** :

  A database is a system of structured data, stored on storage devices to satisfy the information exploitation requirements of many users or many application programs at the same time with different purposes.

3. Database (2)

- **Advantage:**
  - Reduce data duplication to a minimum, ensuring data consistency and integrity.
  - Ensure data is accessible in a variety of ways.
  - Ability to share information with <u>many</u> people, many different applications.

3. Database

- Issues to be addressed:
  - Data sovereignty (chủ quyền dữ liệu)
  - Security and user rights to access information
  - Data disputes
  - Ensure data in case of incident

4. Users

- Database users who are not specialized in the field of information technology and databases → need tools so they can exploit the database when needed.
- IT specialists build applications to serve management purposes.
- Database administrators: database organization, security, authorization, data backup, recovery, data dispute resolution...

5. Database management system (1)

- Database Management System (DBMS) is a system of software that actively supports analysts, designers and database operators.
- Popular DBMS: Microsoft Access, SQL Server, DB2, Oracle, PostgreSQL... most of the current DBMS are based on the relational model.

5. Database management system (2)

- A DBMS must have:
  - Language of communication between user and database
  - Data Dictionary
  - Security measures available upon request
  - Data dispute resolution mechanism
  - Has backup and restore mechanism
  - Ensure independence between data and program

5. Database management system (3)

Languages:

- *Data Description Language (DDL)*: allows declaring database structure, data relationships, rules, and data constraints.
- *Data Manipulation Language (DML)*: allows adding, deleting, and editing data.
- *Structural Query Language (SQL)*: allows the user to query the necessary information.
- *Data Control Language (DCL)*: allows changing table structure, declaring security, and granting permissions to users.

6. Levels of representation of a database

- **Internal level** : (physical level) is the database storage level. What problem needs to be solved? What data? How to store? Where? What indexes are needed? Sequential or random access. For administrators and professional users.
- **Conceptual level** : (Conception or Logical) How many types of data need to be stored? What data? Relationships
- **External level** : of users and application programs

6. Levels of representation of a database